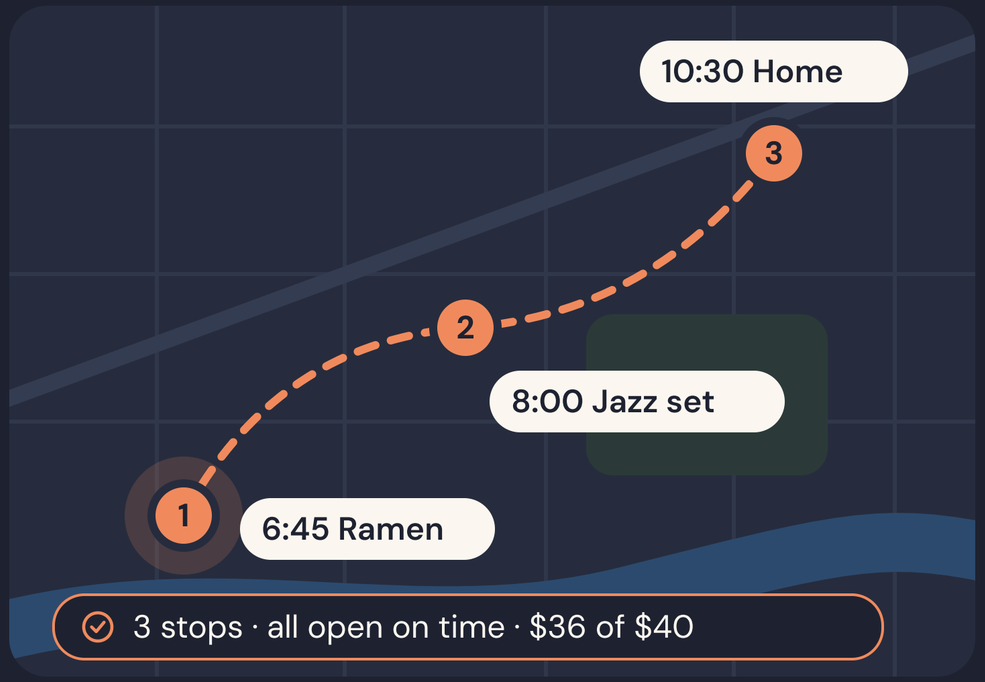
Offhours

An AI agent for free time: live picks for places, concerts and events near you, planned into your calendar.

Who it's for: busy students and young professionals, in any city, with free time in irregular pieces.

Core question: does an LLM paired with a constraint solver plan better than an LLM alone?



Scattered
Ideas live across maps, event sites, feeds and group chats.

Infeasible
Suggestions ignore hours, showtimes, travel time and budget.

Forgotten
Plans miss the calendar, and every search starts from zero.

From one message to a feasible plan

Free from 6:30. Live music under $40, home by 11?

Live search · 3 feasible plans
1. Ramen 6:45 → jazz 8:00
2. Indie show 7:30 → tacos 10:15
3. Comedy 8:00 → dessert

Plan 1, please.

Added to your calendar.

TONIGHT'S PLAN · SOLVED

Ramen, then a jazz set, home by 10:30



- Ramen open from 5 pm
- Jazz at 8:00, 6-min walk
- $36 of a $40 budget
- Home 10:30, before 11

1. **Ask** — Mood, budget, company
2. **Search live** — Places, concerts, events
3. **Solve** — Hours, showtimes, travel
4. **Commit** — Confirm, then calendar
5. **Learn** — Ratings refine picks

The LLM interprets, a solver plans

1 · Request

Free from 6:30. Live music under $40, home by 11?

Plain language in: mood, budget, time, company.

LLM →

2 · Constraints

Window	**6:30–11 pm**
Budget	**≤ $40**
Must	**live music**
Also	**food nearby**
Near	**you, now**

search →

3 · Candidates

- Jazz · 8:00 · $20
- Indie · 7:30 · $25
- Comedy · 8:00 · $15
- Ramen · till 10 pm

+ dozens more records

solver →

4 · Solved plans

1 Ramen → Jazz



2 Indie → Tacos



3 Comedy → Dessert



Orienteering with time windows, in OR-Tools

Then: verify each plan against its sources → LLM explains → you confirm → calendar

Google Calendar · Google Places · Ticketmaster · Weather + travel time · SQLite memory · OR-Tools

Python + FastAPI backend · LLM API with tool calling · React chat UI · recorded API data for evaluation

Any city, one calendar, the full loop

Core build

- Any city; tested in Boston
- Up to 3 stops per plan
- Now + weekend modes
- Google Calendar
- One user, learned preferences

Stretch

- Group plans
- Small-gig web search

Out of scope

- Buying tickets
- Payments
- Mobile app

RISKS AND MITIGATIONS

Invented or stale options

Only search results enter the solver; a verifier checks each plan.

No feasible plan, or a slow solve

Cap stops and candidates; relax soft limits and say what didn't fit.

Thin or costly live data

Blend sources, fall back to evergreen spots, cache and cap API calls.

Calendar writes and location

Confirm every write, with undo; store location only as a neighborhood.

Milestones and how I'll show it works

Oct 29 · Proposal

Scope approved

Feedback on direction before building.

Midpoint

Loop + solver

Chat, live search, solver-built plans, confirmed calendar event.

Final

Benchmark + learning

Hybrid vs. LLM-only results, preference learning in simulation, small pilot.

SUCCESS CRITERIA · 60 SCRIPTED REQUESTS, RECORDED API DATA

≥ 95%

Feasible plans

meet hours, showtimes, travel and budget; reported beside an LLM-only baseline

100%

Grounded

of shown options trace to a live search record

0

Unconfirmed writes

and no duplicate events under injected API failures

< 10 s

Plan latency

median from request to 3 plans, solver included